Experiment 10: Write a python program to implement A* Algorithm

Aim:

To write a Python program to implement the A* search algorithm to find the shortest path from a start node to a goal node.

Algorithm:

1. Initialize the start node.
2. Calculate \(f(n)=g(n)+h(n)\).
3. Select the node with the lowest \(f(n)\).
4. Update the cost of its neighboring nodes.
5. Repeat until the goal node is reached.

Program:

```
graph = {
   'A': {'B': 1, 'C': 4},
   'B': {'D': 2, 'E': 5},
   'C': {'E': 1},
   'D': {'G': 3},
   'E': {'G': 2},
   'G': {}
}
h = {'A': 5, 'B': 4, 'C': 2, 'D': 3, 'E': 1, 'G': 0}
open_list = [(h['A'], 0, 'A', ['A'])]
visited = set()
while open_list:
   open_list.sort()
   f, g, node, path = open_list.pop(0)
   if node == 'G':
      print("Path:", path)
      print("Cost:", g)
      break
```

```
    if node not in visited:
        visited.add(node)
        for next_node, cost in graph[node].items():
            if next_node not in visited:
                new_g = g + cost
                new_f = new_g + h[next_node]
                open_list.append((new_f, new_g, next_node, path + [next_node]))
```

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```
graph = {
    'A': {'B': 1, 'C': 4},
    'B': {'D': 2, 'E': 5},
    'C': {'E': 1},
    'D': {'G': 3},
    'E': {'G': 2},
    'G': {}
}

h = {'A': 5, 'B': 4, 'C': 2, 'D': 3, 'E': 1, 'G': 0}

open_list = [(h['A'], 0, 'A', ['A'])]
visited = set()

while open_list:
    open_list.sort()
    f, g, node, path = open_list.pop(0)

    if node == 'G':
        print("Path:", path)
        print("Cost:", g)
        break

    if node not in visited:
        visited.add(node)

        for next_node, cost in graph[node].items():
            if next_node not in visited:
                new_g = g + cost
                new_f = new_g + h[next_node]
                open_list.append((new_f, new_g, next_node, path + [next_node]))
```

Output:

```
============ RESTART: C:/Users/yadal/OneDrive/Documents/AI/exp 10 a ============
Path: ['A', 'B', 'D', 'G']
Cost: 6
```

Result:

Thus, the Python program to implement the A* search algorithm was executed successfully, finding the path from A to G with a minimum cost of 6.